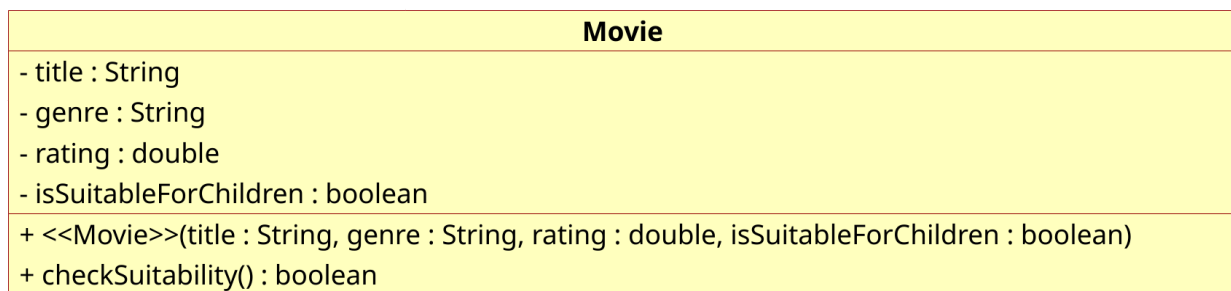


Muhammad Athar Abbas
SP25-BSE-082
Object-Oriented Programming
Lab Assignment # 01

Task # 1:

Movie Class:

UML Diagram:



Program:

```
public class Movie{
    private String title;
    private String genre;
    private double rating;
    private boolean isSuitableForChildren;

    Movie(){};
    Movie(String title , String genre , double rating , boolean
isSuitableForChildren)
    {
        this.title = title;
        this.genre = genre;
        this.rating = rating;
        this.isSuitableForChildren = isSuitableForChildren;
    }
    public boolean checkSuitability(){
        if(rating <= 8.0) {
            return true;
        } else {
```

```

        return false;
    }
}

public void displayInfo(){
    String suited = isSuitableForChildren == true ? "Yes" : "No";
    System.out.println("Movie: \" + title + "\", Genre: \" + genre + \",
Rating: \" + rating + \", Suitable for children: \" + suited);

}

public static void main(String args[]){
    Movie theGodfather = new Movie("The Godfather" , "Crime thriller" ,
9.0 , false);
    Movie theWildRobot = new Movie("The Wild Robot" , "Animation" , 8.5
, true);
    theGodfather.displayInfo();
    theWildRobot.displayInfo();
}
};

```

Output:

```

[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ java Movie
Movie: "The Godfather", Genre: Crime thriller, Rating: 9.0, Suitable for children: No
Movie: "The Wild Robot", Genre: Animation, Rating: 8.5, Suitable for children: Yes
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ █

```

Task # 2:

BankAccount Class:

UML Diagram:

BankAccount
- accountNumber : int - accountHolder : String - balance : double - <u>accountCounter : int</u>
+ <<BankAccount>>(accountNumber : int, accountHolder : String, balance : double) + deposit(amount : double) + withdraw(amount : double) + <u>getTotalAccounts()</u>

Program:

```
public class BankAccount{
    private int accountNumber;
    private String accountHolder;
    private double balance;
    private static int accountCounter;
    BankAccount(){
        accountCounter++;
    };
    BankAccount(int accountNum , String holder , double bal){
        accountNumber = accountNum;
        accountHolder = holder;
        balance = bal;
        accountCounter++;
    }
    public void deposit(double amount){
        showAccountStatus();
        balance += amount;
        System.out.println(amount + " PKR has been deposited in your account.
Your current Account Status: ");
        showAccountStatus();
    }
    public void withdraw(double amount){
```

```

        if (balance - amount < 1000) {
            System.out.println("You cannot withdraw any amount. Minimum balance
will be reached.");
            showAccountStatus();
            return;
        } else if (amount >= balance) {
            System.out.println("You cannot withdraw amount. Insufficient
funds.");
            showAccountStatus();
            return;
        }
        showAccountStatus();
        balance -= amount;
        System.out.println(amount + " PKR has been withdrawn from your account.
Your current Account Status: ");
        showAccountStatus();
    }

    public static void getTotalAccounts() {
        System.out.println("Total number of accounts created : " +
accountCounter);
    }

    public void showAccountStatus() {
        System.out.println("Account Number: " + accountNumber + ", Account
Holder: " + accountHolder + ", Balance: " + balance + " PKR" );
    }

    public static void main(String args[]) {
        BankAccount a1 = new BankAccount(1,"athar",1000);
        BankAccount a2 = new BankAccount(2,"ahmad", 2000);
        BankAccount a3 = new BankAccount(3,"moaz" , 3000);
        a1.deposit(1300);
        a1.withdraw(1100);
        a2.deposit(1400);
        a2.withdraw(1600);
        a3.deposit(300);
        a3.withdraw(2500);
        a3.getTotalAccounts();
    }
}

```

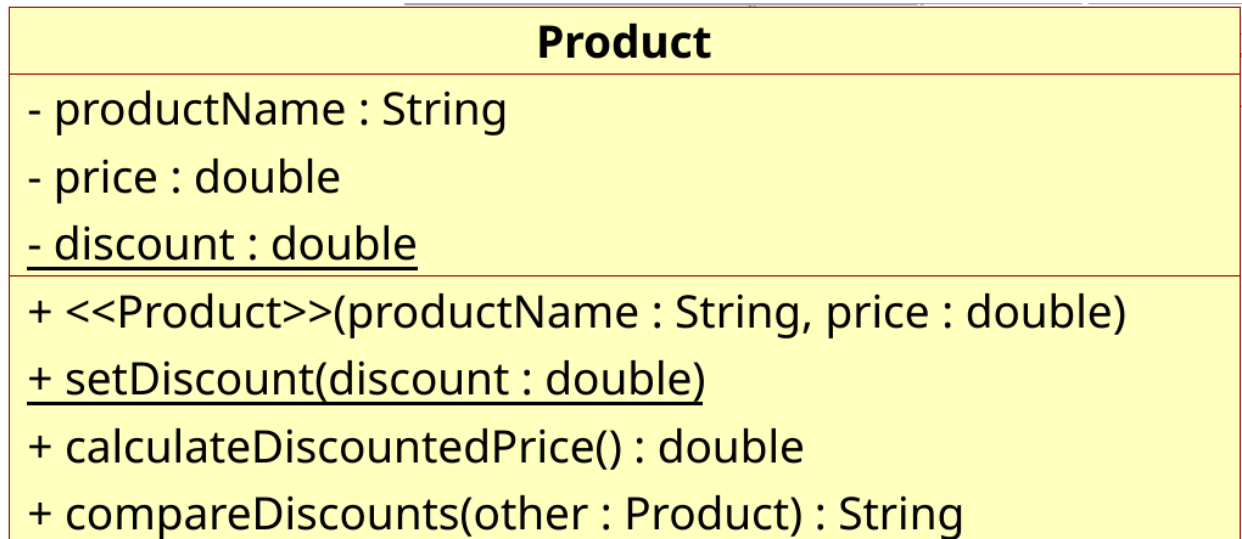
Output:

```
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ javac BankAccount.java
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ java BankAccount
Account Number: 1, Account Holder: athar, Balance: 1000.0 PKR
1300.0 PKR has been deposited in your account. Your current Account Status:
Account Number: 1, Account Holder: athar, Balance: 2300.0 PKR
Account Number: 1, Account Holder: athar, Balance: 2300.0 PKR
1100.0 PKR has been withdrawn from your account. Your current Account Status:
Account Number: 1, Account Holder: athar, Balance: 1200.0 PKR
Account Number: 2, Account Holder: ahmad, Balance: 2000.0 PKR
1400.0 PKR has been deposited in your account. Your current Account Status:
Account Number: 2, Account Holder: ahmad, Balance: 3400.0 PKR
Account Number: 2, Account Holder: ahmad, Balance: 3400.0 PKR
1600.0 PKR has been withdrawn from your account. Your current Account Status:
Account Number: 2, Account Holder: ahmad, Balance: 1800.0 PKR
Account Number: 3, Account Holder: moaz, Balance: 3000.0 PKR
300.0 PKR has been deposited in your account. Your current Account Status:
Account Number: 3, Account Holder: moaz, Balance: 3300.0 PKR
You cannot withdraw any amount. Minimum balance will be reached.
Account Number: 3, Account Holder: moaz, Balance: 3300.0 PKR
Total number of accounts created : 3
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$
```

Task # 3:

Product Class:

UML Diagram:



Program:

```
public class Product{
    private String productName;
    private double price;
    private static double discount;

    Product(){};
    Product(String name , double value) {
        productName = name;
        price = value;
    }
    public static void setDiscount(double sDiscount){
        discount = sDiscount;
    }
    public double calculateDiscountedPrice() {
        double discountAmount = price * (discount / 100);
        double discountedPrice = price - discountAmount;
        return discountedPrice;
    }
}
```

```

    public void compareDiscounts(Product compare) {
        Product higherDiscountProduct = this.calculateDiscountedPrice() >
compare.calculateDiscountedPrice() ? this : compare;
        System.out.println("Product " + higherDiscountProduct.productName +
" has higher discounted price.");
    }

    public void displayInfo() {
        System.out.println("Product: " + productName + ", Original Price: "
+ price + ", Discounted Price: " + calculateDiscountedPrice());
    }

    public static void main(String args[]) {
        Product waterPipe = new Product("Water Pipe" , 2300);
        Product ceilingFan = new Product("Ceiling Fan" , 6700);
        Product.setDiscount(5);
        waterPipe.displayInfo();
        ceilingFan.displayInfo();
        System.out.println("Comparing two products and displaying the one
with higher price ");
        ceilingFan.compareDiscounts(waterPipe);
    }
}

```

Output:

```

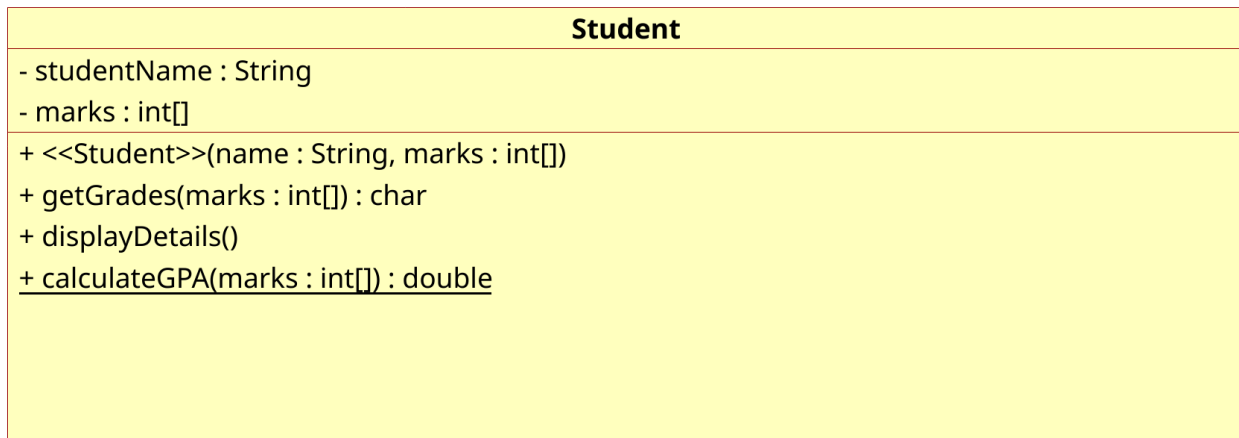
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ java Product
Product: Water Pipe, Original Price: 2300.0, Discounted Price: 2185.0
Product: Ceiling Fan, Original Price: 6700.0, Discounted Price: 6365.0
Comparing two products and displaying the one with higher price
Product Ceiling Fan has higher discounted price.
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ █

```

Task # 4:

Student Class:

UML Diagram:



Program:

```
import java.util.Scanner;

public class Student{
    private String studentName;
    private int[] marks;
    public Student(){};
    public Student(String name , int[] numbers){
        studentName = name;
        marks = numbers;
    }
    public static double calculateGPA(int[] numbers){
        int sum = 0;
        for (int value : numbers){
            sum += value;
        }
        double marksAvg = sum / numbers.length;
        double gpa = marksAvg / 25;
        if (gpa > 4.0) {
            return 4.0;
        }
        return gpa;
    }
}
```



```

}

public char[] getGrades() {
    char[] gradesArray = new char[marks.length];
    for (int i = 0; i <= marks.length - 1 ; i++) {
        int value = marks[i];
        char grade;
        if (value >= 90) {
            grade = 'A';
        } else if (value >= 80) {
            grade = 'B';
        } else if (value >= 70) {
            grade = 'C';
        } else if (value >= 60) {
            grade = 'D';
        } else {
            grade = 'F';
        }
        gradesArray[i] = grade;
    }
    return gradesArray;
}

public void displayDetails() {
    int size = marks.length;
    char[] grades = getGrades();
    String gradeList = "[";
    for (int i = 0 ; i <= size - 1 ; i++) {
        gradeList += grades[i];
        if (i != size - 1) {
            gradeList += ", ";
        }
    }
    gradeList += "]";
    System.out.println("Student: " + studentName + ", GPA: " +
        calculateGPA(marks) + ", Grades: " + gradeList);
}

public static void main(String args[]) {
    System.out.print("Enter the number of students: ");
    Scanner input = new Scanner(System.in);
    int numOfStudents = input.nextInt();
}

```

```

input.nextLine();
Student[] stdArray = new Student[numOfStudents];
for (int i = 0 ; i < numOfStudents ; i++ ) {
    System.out.print("Name : ");
    String name = input.nextLine();
    System.out.print("Number of Subjects: ");
    int subs = input.nextInt();
    input.nextLine();
    int[] marksArray = new int[subs] ;
    for (int j = 0 ; j < subs; j++) {
        System.out.print("Enter marks: ");
        marksArray[j] = input.nextInt();
        input.nextLine();
    }
    Student std = new Student(name , marksArray);
    stdArray[i] = std;
}
System.out.println("=====");
Student topPerformer = null;
String topPerformerName = "";
double minGPA = 0.0;
for (Student std : stdArray) {
    std.displayDetails();
    double currentGPA = std.calculateGPA(std.marks);
    if (currentGPA > minGPA) {
        topPerformer = std;
        minGPA = currentGPA;
    }
};
System.out.println("Top Performer: " + topPerformer.studentName);
};
}

```

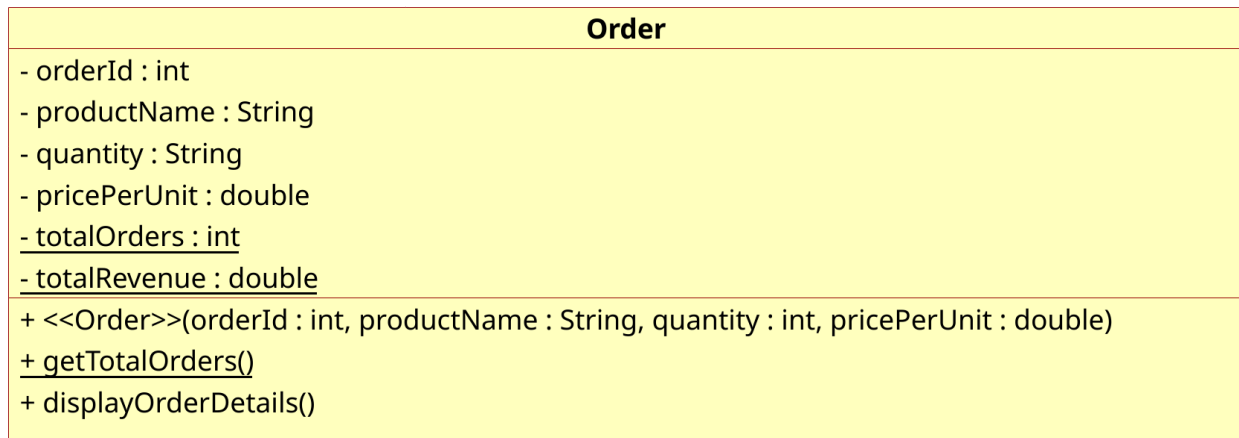
Output:

```
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ javac Student.java
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ java Student
Enter the number of students: 3
Name : Athar Abbas
Number of Subjects: 3
Enter marks: 76
Enter marks: 84
Enter marks: 92
Name : Jamal Nazir
Number of Subjects: 3
Enter marks: 55
Enter marks: 56
Enter marks: 57
Name : Himat Javed
Number of Subjects: 3
Enter marks: 23
Enter marks: 89
Enter marks: 75
=====
Student: Athar Abbas, GPA: 3.36, Grades: [C, B, A]
Student: Jamal Nazir, GPA: 2.24, Grades: [F, F, F]
Student: Himat Javed, GPA: 2.48, Grades: [F, B, C]
Top Performer: Athar Abbas
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$
```

Task # 5:

Order Class:

UML Diagram:



Program:

```
public class Order{
    private int orderId;
    private String productName;
    private int quantity;
    private double pricePerUnit;
    private static int totalOrders;
    private static double totalRevenue;
    public Order() {
        totalOrders++;
        totalRevenue += quantity * pricePerUnit;
    };
    public Order(int id, String name, int qty, double unitPrice){
        orderId = id;
        productName = name;
        quantity = qty;
        pricePerUnit = unitPrice;
        totalOrders++;
        totalRevenue += quantity * pricePerUnit;
    }
    public static void getTotalOrders(){
```

```

        System.out.println("Total Orders: " + totalOrders);
    }
    public void displayOrderDetails(){
        String orderDetails = "Order Id: " + orderId + ", Product: " +
productName + ", Quantity: " + quantity + ", Total Price: " + quantity *
pricePerUnit + " PKR";
        if (quantity >= 50) {
            double totalPrice = quantity * pricePerUnit;
            double discount = totalPrice * 0.10;
            double discountedPrice = totalPrice - discount;
            orderDetails += " Total Price after Discount: " + discountedPrice;
        }
        System.out.println(orderDetails);
    }
    public static void main(String args[]) {
        Order o1 = new Order(111 , "cup", 20 , 140);
        o1.displayOrderDetails();
        Order o2 = new Order(321 , "honey" , 55 , 245);
        o2.displayOrderDetails();
        Order o3 = new Order(456 , "jam" , 40, 350);
        o3.displayOrderDetails();
        System.out.println("Total Orders: " + Order.totalOrders);
        System.out.println("Total Revenue: " + Order.totalRevenue);
    }
}

```

Output:

```

[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ javac Order.java
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ java Order
Order Id: 111, Product: cup, Quantity: 20, Total Price: 2800.0 PKR
Order Id: 321, Product: honey, Quantity: 55, Total Price: 13475.0 PKR Total Price after Discount: 12127.5
Order Id: 456, Product: jam, Quantity: 40, Total Price: 14000.0 PKR
Total Orders: 3
Total Revenue: 30275.0
[athar@athar ~/semester2/OOP/java_programs/drivelabs/lab3]$ █

```